

CLASH OF THE **CLAWS**

RVCE_Null Pointers Your Autonomous Overnight Engineer for Tech Debt

**Ayush Aman
Astitwa Tanmay
Varun Dhandharia
Animesh Sapra**



The Problem With Tech Debt

Developers drown in flagged-but-never-fixed issues every single day



Linters Only Flag, Never Fix

Ruff, ESLint, Pylint identify hundreds of issues per run — but every fix still needs a developer to open the file manually. The flag is raised; the work remains.



Security Scanners Create Alert Fatigue

pip-audit and Snyk report vulnerable dependencies every day, but never bump versions or open PRs automatically. Teams stop reading alerts after week one.



No Overnight Automation Exists

There is no tool that runs at 2 AM, patches real issues, validates fixes in isolation, and opens a clean PR by morning — developers arrive to the same debt they left behind.

23%

of engineer time lost to tech debt

4×

slower feature delivery over time

\$85B

lost annually to poor code quality

0

tools that fix + validate + PR automatically

What Exists — And What's Missing

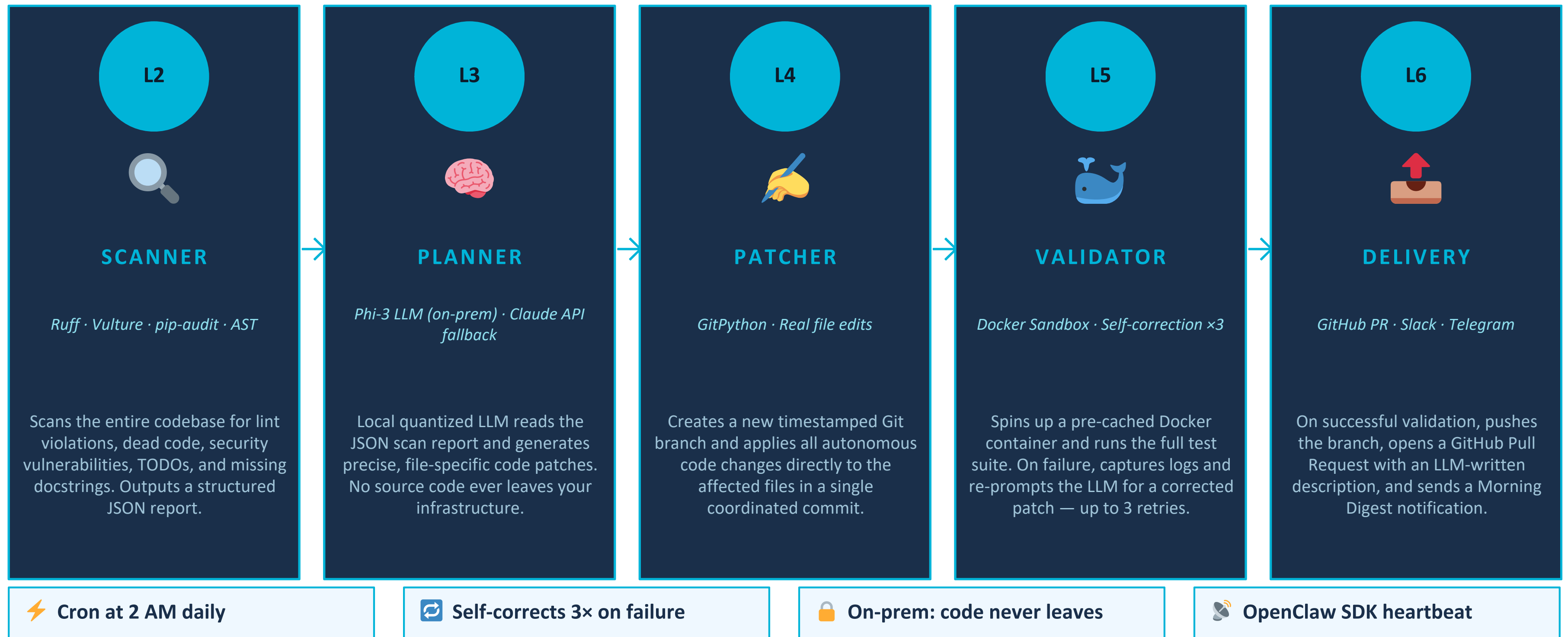
The market has detection tools. Ghost Committer is the only complete fix-and-deliver loop.

Capability	Linters (Ruff/ESLint)	Security (pip-audit)	Copilot Bots	Ghost Committer ✓
Detect lint issues	✓	✗	✓	✓
Detect dead code	✓	✗	✗	✓
Scan vulnerabilities	✗	✓	✗	✓
Auto-fix code	✗	✗	✗	✓
Validate fix (run tests)	✗	✗	✗	✓
Self-correct on failure	✗	✗	✗	✓
Open GitHub PR	✗	✗	✗	✓
Runs overnight autonomously	✗	✗	✗	✓

Ghost Committer is the ONLY solution that closes the full loop: Detect → Fix → Validate → PR — automatically.

Ghost Committer — 6-Layer Autonomous Pipeline

An AI agent that scans, fixes, validates, and delivers PRs while your team sleeps



Live End-to-End Pipeline Walkthrough

- 1 Trigger: Cron at 2 AM** scheduler.py fires python main.py . — or run immediately with python main.py <repo_path>
- 2 Scanner Fires All Checks** Ruff, Vulture, pip-audit + custom AST scan execute in sequence. Output: structured JSON report with all findings.
- 3 Planner Generates Patches** Phi-3 LLM (running locally via llama.cpp) reads the JSON findings and generates file-specific code patches for each issue.
- 4 Patcher Edits Real Files** New branch created: chore/ghost-auto-YYYYMMDD-HHMM. All autonomous fixes committed in a single clean commit.
- 5 Docker Validator Runs Tests** Pre-cached ghost-validator:latest container spins up. Tests run in isolation. Failure? LLM re-prompted with full error trace.
- 6 PR + Morning Digest Sent** GitHub PR opened with LLM-generated description. Telegram bot and Slack workspace notified with stats at wake-up.

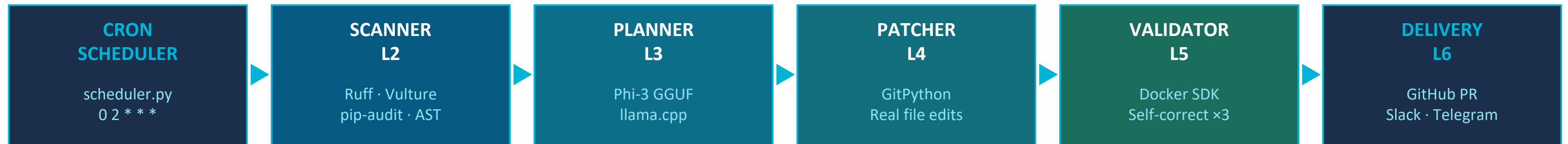
```
... terminal — python main.py .

--- Waking up Ghost Committer ---

[OpenClaw] Sending status: starting
[Scanner] Ruff: 8 lint issues found
[Scanner] Vulture: 2 dead code items
[Scanner] pip-audit: 1 CVE detected
[Planner] Phi-3 generating patches...
[Patcher] Branch: chore/ghost-auto-...
[Validator] Docker sandbox spinning up
[Validator] Tests: PASS ✓ (45 sec)
[Delivery] PR #42 opened on GitHub
[Delivery] Telegram digest sent ✓

--- Ghost Committer: SUCCESS ---
```


Technical Architecture & Full Stack



Static Analysis

- Ruff — Python linting (PEP8, unused imports, style)
- Vulture — Dead code detection (unused functions, vars)
- pip-audit — CVE / security vulnerability scanning
- Custom AST — TODO tracker, cyclomatic complexity, docstring checker

AI / LLM Layer

- Phi-3-mini GGUF (primary) — runs fully on-prem via llama-cpp-python
- n_ctx=4096 — full file context for accurate patching
- Claude API (anthropic SDK) — opt-in cloud fallback
- Strict Python prompt templates with exponential backoff retry

Infrastructure

- Docker SDK — sandbox test execution with pre-cached image
- GitPython + PyGithub — branch ops, commit, PR creation
- FastAPI + Uvicorn — internal REST API for status
- Slack SDK + Telegram Bot API — morning digest notifications

Platform (OpenClaw)

- openclaw.yaml: 2 CPU · 4Gi RAM · cron 0 2 * * *
- OpenClaw SDK heartbeat: starting → success / failed
- Secret injection: GITHUB_TOKEN, OPENCLAW_API_KEY, SLACK_BOT_TOKEN
- Status broadcast to OpenClaw dashboard for visibility

Real-World Impact & Use Cases

90%**Faster builds**

Pre-cached Docker image

3×**Auto-retries**

Self-correction loop

0h**Human hours**

Overnight automation

6**Layer pipeline**

Scan → PR delivery



Startup Engineering Team

5-person startup — shipping daily

- PR opens each morning fixing 8 lint violations
- Vulnerable dependency auto-patched overnight
- 12 new docstrings generated autonomously
- Zero engineer hours spent on debt cleanup



Open Source Maintainer

Solo maintainer, 200+ GitHub issues

- Auto-resolves all code quality PRs overnight
- Frees maintainer time for feature development
- New contributors find a consistently clean codebase
- PRs auto-link back to related GitHub issues



Enterprise Compliance

Security mandate: 0 critical CVEs allowed

- Auto-bumps vulnerable packages in requirements.txt
- PR opened and ready before morning standup
- Full audit trail via Git commit history
- OpenClaw dashboard gives fleet-wide visibility

**Scale Potential: From solo repos → enterprise fleets with multi-repo OpenClaw orchestration and per-team config dashboards**

Why Ghost Committer Wins — Our Moats

Every other tool stops at detection. We complete the entire loop.

01 Self-Healing Intelligence



The agent reads its own pytest failure logs, re-prompts the Phi-3 LLM with the exact error traceback, and retries a fresh patch. This autonomous feedback loop — unique in open-source tooling — means the agent fixes its own mistakes without any human involvement. Up to 3 retries before gracefully reporting failure.

02 True On-Prem Privacy



Runs entirely on-device using Phi-3-mini GGUF via llama.cpp (n_ctx=4096). Source code never leaves the organization's servers. Claude API is available only as an opt-in cloud fallback. Perfect for enterprise IP-sensitive and regulated codebases.

03 90% Faster Validation (45 sec builds)



Custom ghost-validator:latest Docker image pre-caches all system dependencies. First run takes ~10 minutes to build. All subsequent nightly runs complete in ~45 seconds. This makes overnight automation actually practical and scalable at enterprise scale.

04 Full-Loop Automation — Scan → PR



Every other tool stops at detection or suggestion. Ghost Committer completes the entire 6-step loop: Scan → Plan → Patch → Validate → Commit → PR → Notify. Nothing else on the open-source market does all six steps autonomously in a single overnight run.

Demo Links & Project Submission



GitHub Repo

github.com/NullPointers-RVCE/Ghost-Committer

Full source, README, and openclaw.yaml



Test Bench Repo

github.com/Astitwa73/ghost-test-bench

Intentionally debt-laden repo used for live demo



Telegram Bot

[@GhostCommitterBot](https://t.me/GhostCommitterBot)

Live morning digest bot — fully operational



Project Status

FULLY OPERATIONAL

All 6 layers running as of May 7, 2026

⚡ Quick Start

```
git clone github.com/NullPointers-RVCE/Ghost-Committer
```

```
python -m venv .venv
```

```
pip install -r requirements.txt
```

```
cp .env.example .env
```

```
# Set GITHUB_TOKEN
```

```
# Set OPENCLAW_API_KEY
```

```
# Set SLACK_BOT_TOKEN
```

```
# Run immediately on current repo:
```

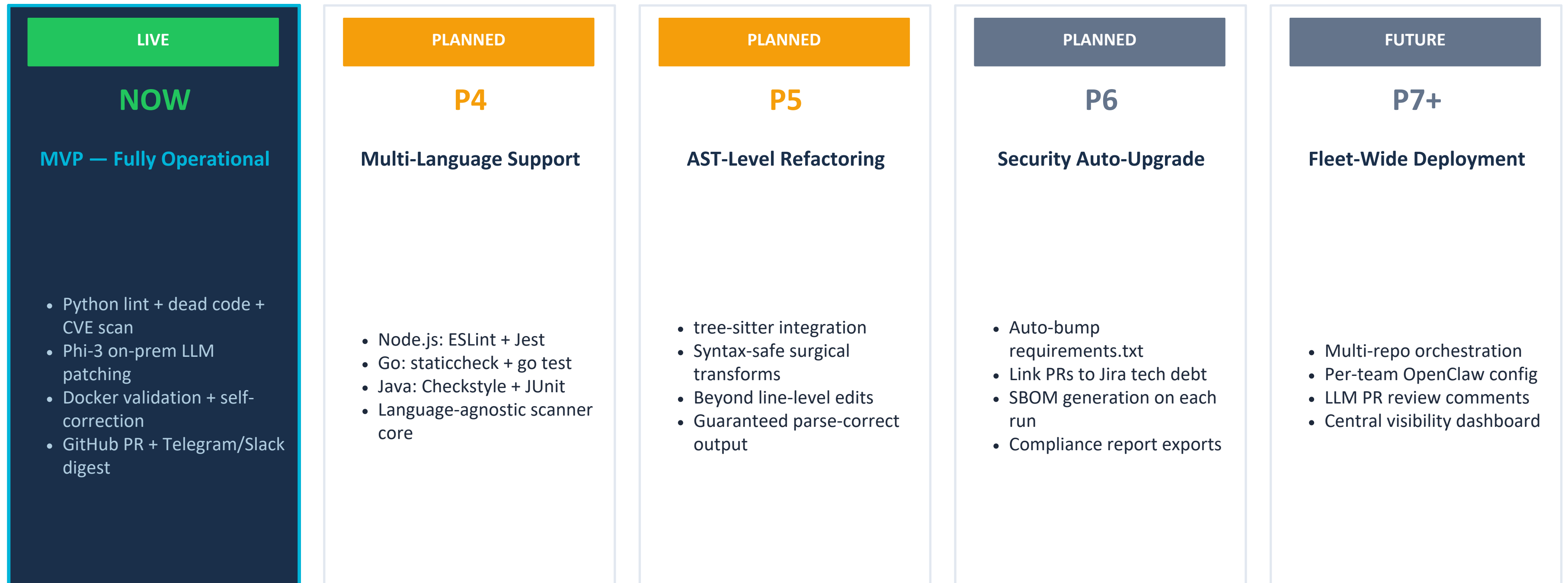
```
python main.py .
```

```
# Start 2AM nightly scheduler:
```

```
python scheduler.py
```

Roadmap — Beyond the Hackathon

Ghost Committer has a clear product trajectory from hackathon MVP to enterprise platform



Phase 8 Vision: Fleet-wide OpenClaw deployment — multi-repo orchestration with per-team config and a central monitoring dashboard for engineering leads